FireNav: A Physics-Grounded Benchmark for Cooperative Multi-Agent Navigation in Dynamic Indoor Fire Environments

Anonymous Author(s)

Abstract

Multi-agent robotic systems offer a promising approach to search and rescue in indoor fires, but their systematic design and evaluation remain difficult. Real-world fire experiments are dangerous, expensive, and difficult to carry out, while existing embodied navigation benchmarks primarily target single-agent tasks in static or quasi-static, hazard-free environments. To fill in this gap, we present FireNav, the first physics-grounded benchmark for cooperative multi-agent navigation in dynamic indoor fire environments. Its underlying infrastructure, FireWorld, transforms static 3D scenes into dynamic fire worlds through semantic scenario generation, physics-inspired voxel hazard propagation, synchronized rendering, and multimodal sensor data synthesis. To support human search-and-rescue, we introduce a category-extensible task construction pipeline that enables externally instantiated targets while preserving the original navigation interface. Building on this infrastructure, FireNav provides a unified hazard-aware navigation framework integrating multimodal perception, online hazard estimation, cooperative global planning, and local risk-aware navigation. Extensive evaluations show that dynamic fire reduces the Success Rate (SR) and Success weighted by Path Length (SPL) of conventional navigation by 39.1% and 31.2%, respectively. In comparison, FireNav improves SR and SPL by 53.0% and 35.7% on average under the same fire conditions, while reducing Average Hazard Exposure (AHE) and Severe Hazard Exposure Rate (SHER) by 15.0% and 20.5%. Experiments across diverse scenes, fire conditions, and team sizes further demonstrate its robustness, while deployment on two physical quadruped robots shows the practical feasibility of transferring the framework to real-world cooperative search. Dataset, code, and demonstrations are available at https://firenav.github.io/.

CCS Concepts

• **Computer systems organization → Robotics**; • **Computing methodologies → Multi-agent planning**; • **General and reference** → *Evaluation*; *Measurement*.

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1 INTRODUCTION

Natural and human-made disasters pose immediate threats to life and infrastructure, demanding rapid, reliable, and coordinated emergency response. In hazardous environments such as indoor fires, high temperature, dense smoke, and structural instability, human intervention becomes extremely dangerous. For example, a recent devastating fire in Tai Po, Hong Kong severely impeded human access and situational awareness during the early response phase and became the deadliest fires in Hong Kong in decades [4].

Mobile robots have emerged as a promising alternative for operating in life-threatening environments. By enabling autonomous exploration, hazard perception, and target search under severe perception degradation, robotic systems can provide rapid situational awareness while reducing the risk to human responders and supporting more informed decision-making. However, disaster environments are dynamic and time-critical, making single-robot exploration inefficient. This motivates *multi-agent cooperative search and navigation*, where multiple robots can explore in parallel, share observations, coordinate their movements, and distribute search tasks to improve coverage and reduce response time.

Realizing this potential requires not only effective coordination algorithms, but also a systematic way to develop and evaluate them under representative disaster conditions. However, conducting experiments in real fire environments is inherently dangerous, expensive, and difficult to control or reproduce. Critical factors such as hazard evolution, smoke density, target locations, and sensing conditions cannot be systematically varied while maintaining identical experimental conditions across different methods.

Despite rapid progress in embodied simulation and multi-agent navigation, existing navigation benchmarks primarily focus on single-agent tasks and predominantly static or quasi-static and hazard-free environments [9, 23, 32, 34, 35, 38].

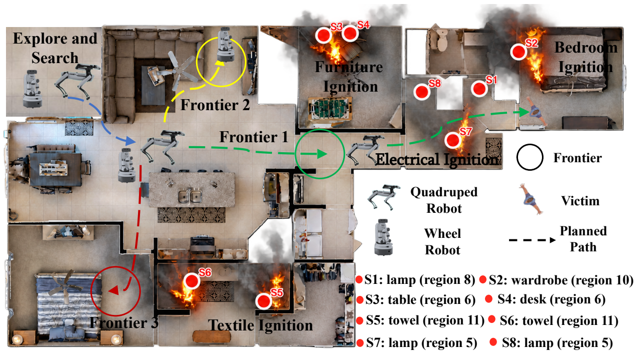

Figure 1: FireNav for multi-agent search and rescue in indoor fire scenarios.

Although some simulation platforms support environment-driven dynamics, disaster scenarios, or multi-agent navigation, these capabilities are typically studied in isolation [2, 13, 41]. As a result, existing benchmarks lack an integrated, physics-grounded framework for scalable fire generation, dynamic hazards, hazard-aware sensing, and cooperative multi-agent navigation. Building such a benchmark, however, raises three key challenges.

A1: How can we construct scalable and reproducible dynamic fire worlds from existing datasets? The challenge is not simply to render flames or manually place hazardous regions, but to automatically transform static 3D scenes into diverse and controllable dynamic fire scenarios while preserving the underlying geometry and semantics. Moreover, this process must scale beyond a small set of handcrafted scenes to large-scale real-world scene datasets, enabling systematic generation of dynamic fire environments.

A2: How can we achieve physics-grounded yet efficient fire hazard simulation while supporting real-time embodied navigation? Fire, smoke, and temperature evolve jointly over space and time and directly affect both robot perception and navigation. A practical benchmark must therefore model these dynamics in a physically plausible yet computationally efficient manner, while enabling low-latency synchronization of the evolving hazard with agent observations and actions.

A3: How should cooperative multi-agent navigation be formulated and evaluated under dynamic fire hazards? Fire environments introduce challenges beyond conventional multi-agent navigation: rapidly changing hazards alter safe and traversable regions; smoke and thermal effects degrade sensor observations; and time-critical decisions must balance task efficiency against hazard exposure.

To address these challenges, we introduce FireNav, the first physics-grounded and scalable benchmark for cooperative multi-agent search and navigation in dynamic indoor fire environments. FireNav defines the standardized tasks, evaluation protocols, baseline methods, and is built on FireWorld, its underlying infrastructure for dynamic-fire simulation.

To address **A1**, FireWorld automatically transforms static indoor scenes into configurable and reproducible fire worlds by leveraging scene semantics and material properties for ignition selection and scenario generation. A category-extensible task-construction pipeline further supports externally instantiated human targets.

To address **A2**, FireWorld employs a lightweight physics-inspired voxel model to simulate the evolution of flame, smoke, and temperature, and couples the resulting hazard fields with multimodal sensor synthesis. Fire propagation is decoupled from the robot-control loop, while down-sampling rendering and GPU-parallelized volume compositing enable low-latency synchronization between evolving hazards and online navigation.

To address **A3**, FireNav builds on FireWorld to provide a unified hazard-aware multi-agent navigation framework. It integrates multimodal perception, online hazard estimation, cooperative global planning, and local risk-aware navigation, and supports systematic evaluation of classical, learning-based, and vision language model (VLM)-based global–local planners across diverse scenes and fire conditions.

Our evaluation yields several key findings. Different planning strategies exhibit distinct performance trade-offs and complementary strengths, while dynamic fire significantly degrades conventional navigation performance, reducing Success Rate (SR) and Success weighted by Path Length (SPL) by 39.1% and 31.2%, respectively. Under the same fire conditions, FireNav improves SR and SPL by 53.0% and 35.7% on average, while reducing Average Hazard Exposure (AHE) and Severe Hazard Exposure Rate (SHER) by 15.0% and 20.5%. These gains highlight the benefits of multimodal perception and hazard-aware planning and remain consistent across diverse scene scales, fire conditions, team sizes. Deployment on two physical quadruped robots further demonstrates the practical transferability of the framework to real-world cooperative search. The dataset, source code, and videos are available at https://firenav.github.io/. The contributions of the paper are as follows:

- We develop FireWorld, a physics-based infrastructure that converts static 3D scenes into scalable dynamic fire environments through semantic fire-plan generation, physics-inspired voxel propagation, synchronized rendering, and fire-aware sensor data synthesis.
- We propose FireNav, the first unified hazard-aware multi-agent navigation framework that integrates multimodal perception, online hazard estimation, cooperative global planning, and local risk-aware navigation for cooperative search under dynamic fire hazards.

- We conduct a comprehensive evaluation across diverse planners, scenes, fire conditions, and team sizes, demonstrating the effectiveness and robustness of multimodal perception and hazard-aware planning under evolving fire hazards. We further validate our framework on two physical quadruped robots performing cooperative target search.

2 BACKGROUND

2.1 Simulation Environment

Habitat-Lab is a widely used platform for embodied AI research, providing configurable agents, sensors, action spaces, and standardized interfaces for navigation tasks in photorealistic 3D environments [23]. Built upon this platform, the Habitat–Matterport 3D (HM3D) dataset provides large-scale reconstructions of real-world indoor environments with diverse layouts and realistic geometry [22]. HM3DSem further augments HM3D with dense semantic annotations at the object-instance level, enabling agents to reason jointly about scene geometry and semantic content [37]. Together, Habitat-Lab and HM3DSem provide realistic indoor geometry and instance-level semantics, providing a foundation for our benchmark.

2.2 Dynamic Fire Simulation

An indoor fire is an autonomous and continuously evolving physical process. Combustion and heat release drive temperature changes, flame growth, buoyant plume transport, smoke generation, and lateral spreading near ceilings and structural boundaries. High-fidelity fire engineering tools such as the NIST Fire Dynamics Simulator (FDS) model these processes using computational fluid dynamics, numerically solving equations for low-speed, thermally driven flow with an emphasis on heat and smoke transport [18]. However, their high computational cost hinders direct integration with robotic simulators and limits scalable and reproducible evaluation.

2.3 Multi-Agent Cooperative Navigation

Multi-agent cooperative navigation extends embodied navigation from a single agent to a team of robots that jointly explore and search an initially unknown environment. By executing actions in parallel and sharing spatial and semantic information, multiple agents can improve environment coverage, reduce search time, and distribute exploration effort across different regions [1, 21, 25, 29, 40]. A typical cooperative navigation pipeline therefore consists of three closely coupled components: shared perception and mapping, global task or frontier assignment, and local navigation toward the assigned targets [2, 14, 39, 40]. The shared map provides a common representation of explored space and semantic observations, while global coordination determines where individual agents should navigate to maximize team-level exploration or search efficiency. Each agent then executes its assigned goal using a local planner while periodically updating the shared representation with newly acquired observations.

3 BENCHMARK INFRASTRUCTURE

In this section, we describe FireWorld, the underlying infrastructure of FireNav for constructing realistic, scalable and category-extensible fire-navigation environments. We instantiate FireWorld on real-world 3D scans from HM3DSem, including 145 training and 36 validation scenes. As illustrated in Fig. 2, FireWorld extends the Habitat–HM3D Object-Goal Navigation setting along three complementary dimensions: (1) dynamic fire world generation, (2) multi-modal sensor simulation, and (3) category-extensible navigation task construction. Specifically, it first transforms static indoor scenes into time-varying fire environments. Then, it couples evolving hazards with sensor observations and degradation models. Finally, it includes a category-extensible ObjectNav construction pipeline that supports both native and externally instantiated target categories under a unified Habitat-compatible protocol. **Notably, the proposed infrastructure construction pipeline is not specific to Habitat–HM3D and can be readily extended to other simulators and navigation datasets that provide semantic scene annotations.**

3.1 FireWorld Construction

3.1.1 Scene Parsing and Physical Property Grounding. The objective of this stage is to construct a spatially grounded scene representation that provides the semantic, geometric, and physical information required for subsequent fire-scenario generation. Given a semantically annotated 3D scene, we build an instance-level inventory O in which each object is associated with its semantic category, 3D extent, floor assignment, and material-dependent fire properties. Specifically, semantic object instances are recovered from the scene annotations by aggregating labeled mesh geometry at the instance level. Each instance is transformed into the simulator coordinate frame, localized using an axis-aligned bounding box, and assigned to the corresponding floor according to its vertical extent. Structural instances, such as walls, floors, and ceilings, are separately rasterized into a voxelized occupancy mask to constrain fire propagation across solid boundaries. Finally, each semantic category is associated with fire-relevant material properties, including flammability and smoke yield, which subsequently condition ignition selection, flame propagation, and smoke generation.

3.1.2 Fire Plan Generation. Given the scene inventory O, FireWorld converts a fire-scenario type c, intensity level q,

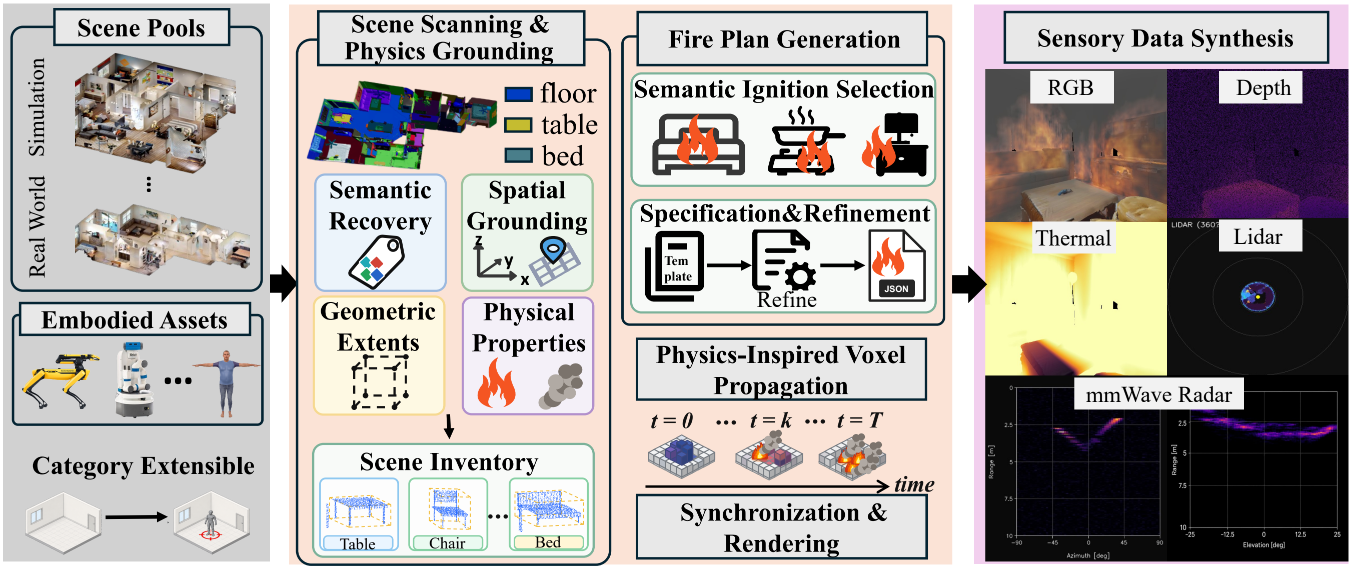

Figure 2: Overview of FireWorld construction pipeline. Starting from a pool of indoor scenes and embodied assets, the system first performs scene scanning to recover semantic, spatial, geometric, and physical attributes and constructs a scene inventory. Based on this inventory, a fire plan is generated through semantic ignition selection, propagation configuration, and episode-level refinement. The resulting plan drives physics-inspired voxel propagation and synchronized rendering, which are finally used to simulate fire-aware multi-modal observations, including RGB, thermal, and radar-based sensing outputs.

and random seed s into a machine-readable fire plan through a semantics- and physics-informed procedure that jointly controls ignition placement, hazard severity, and episode difficulty:

$$\Pi = f_{\text{plan}}\left(\mathcal{O}; c, q, s\right) = \left(\mathcal{I},\ \boldsymbol{\theta}_{\text{prop}}\right), \tag{1}$$

where $\mathcal{I}$ is the set of ignition sources and $\boldsymbol{\theta}_{\text{prop}}$ contains the parameters of the propagation model. The plan therefore specifies *when and where* fires are introduced and *how* the subsequent propagation model is parameterized.

Rather than placing ignition sources uniformly at random, FireWorld uses semantic and material priors to generate physically plausible ignition configurations. Primary ignition objects are selected according to their semantic relevance and material-dependent flammability, while additional sources favor spatially related combustible objects. This allows the same procedure to generate semantically coherent yet diverse fire scenarios across different indoor scenes. We provide four representative fire-scenario templates, each specifying preferred and fallback ignition-object categories. Each template can further be instantiated with one of three intensity presets, which determines a consistent set of source and propagation parameters controlling the evolution of flame, smoke, and temperature. The resulting machine-readable plan can be modified to create reproducible scenario variations without changing the simulation pipeline.

To construct challenging yet solvable navigation scenarios, we optionally refine automatically-generated fire plans at the episode level. For each fixed episode, we preserve the original start and goal, and then select combustible objects near the risk-free shortest path as candidate ignition sources. We search for fire plans that make the nominal shortest path hazardous while keeping a moderately longer, topologically distinct route safe and goal-reachable. Detailed ignition sampling, template configuration, and refinement procedures, together with representative scene visualizations are provided in Appendix A.3.

3.1.3 Physics-Inspired Voxel Propagation. Given a generated fire plan, FireWorld evolves the spatial fire state on a three-dimensional voxel grid using a lightweight physics-inspired propagation model. In particular, the propagation model consists of three stages: (i) heat and smoke transport, (ii) combustion and local flame spread, and (iii) dissipation and source forcing. First, heat diffuses to nearby free-space voxels, while buoyancy drives hot air and smoke upward and induces lateral spreading near ceilings; structural elements constrain transport across solid boundaries. Second, combustible voxels ignite once sufficient fuel and temperature conditions are met, consume fuel, and generate additional heat, flame, and smoke, allowing fire to spread to nearby combustible regions. Finally, away from active sources, flame and smoke decay and temperature relaxes toward ambient conditions,

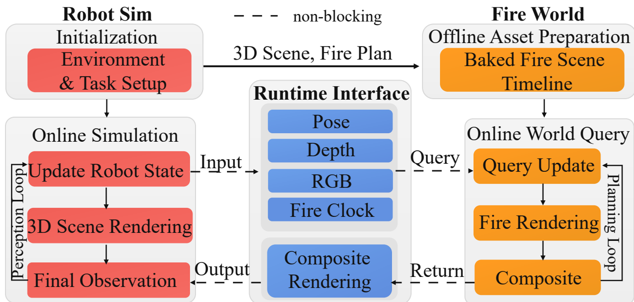

Figure 3: FireWorld synchronization and rendering workflow.

whereas fire-plan ignition sources provide sustained forcing for their prescribed durations. Together, these mechanisms produce spatially structured and time-varying hazard fields while remaining lightweight enough for scalable benchmark generation. Detailed numerical update rules and simulation parameters are provided in Appendix B.

3.1.4 Fire-World Synchronization and Rendering. As illustrated in Fig. 3, FireWorld decouples computationally intensive fire propagation from online robot simulation. During initialization, the 3D scene and fire plan are processed offline to generate a time-indexed fire-scene timeline containing the evolving flame, smoke, and temperature fields. During online navigation, the robot updates its current pose, clean RGB-D observations, along with a synchronized fire clock through a lightweight runtime interface. Based on the fire clock and robot viewpoint, FireWorld queries the corresponding hazard state, renders the visible fire and smoke, and composites the resulting hazard layers with the clean scene rendering to produce the final fire-aware observation. The composite observation is then returned to the robot simulator for subsequent perception and planning, forming a closed interaction loop between the robot simulation and FireWorld.

For online hazard rendering, FireWorld adopts forward volume rendering over the retrieved 3D hazard state. Camera rays are marched through the hazard volume and composited front-to-back using an emission–absorption model [17], where smoke attenuates the background scene and flame contributes self-emissive radiance. To reduce rendering cost, ray marching is performed at a downsampled resolution and the resulting hazard layers are upsampled before compositing. To improve visual realism without altering the underlying hazard state, we further apply rendering-only flame appearance controls, including temporal flickering, boundary irregularity, color variation, and animation speed [12, 19]. The ray-marching and compositing operations are parallelized on the GPU, enabling efficient per-pixel hazard sampling and blending. Together, downsampled rendering and GPU acceleration enable low-latency generation of fire-aware observations (detailed measurements are reported in Appendix A.5).

3.2 Fire Sensor Data Synthesis

As discussed in Appendix C.1, sensing modalities exhibit complementary behavior under fire and smoke. Accordingly, FireWorld adopts RGB-D, thermal imaging, and mmWave radar for fire-aware navigation: RGB-D provides semantic and geometric information under nominal conditions, while thermal and mmWave sensing function when smoke degrades visible-spectrum imaging and optical ranging.

All modalities are synthesized from the same synchronized fire state, ensuring spatially and temporally consistent observations. RGB images combine the clean simulator view with the volumetrically rendered smoke and flame fields described in Sec. 3.1.4. Depth observations are progressively degraded with increasing smoke density through range-dependent noise, invalid measurements, and dropout, reflecting the reduced reliability of optical ranging in fire-smoke environments [27]. Thermal images are generated by projecting the simulated temperature field into the camera view and mapping it to the sensor image domain with bounded noise and dynamic-range normalization. Finally, mmWave radar is simulated as a smoke-resilient geometric sensing modality. We first construct a clean LiDAR-like point cloud by stitching depth observations from multiple yaw-oriented depth cameras to provide a 360° view of the surrounding scene. This dense geometric point cloud is then spatially and angularly downsampled to approximate the angular resolution and point density of a single-chip mmWave radar within a predefined radar field of view, yielding a radar-like point-cloud observation. The resulting representation preserves the underlying scene geometry while reflecting the characteristic sparsity of mmWave sensing [15, 20].

By grounding RGB-D degradation, thermal response, and mmWave robustness in empirical sensor behavior, FireWorld provides a consistent and physics-grounded observation model for evaluating perception and navigation under dynamic fire conditions. Representative examples of the simulated sensor observations are provided in Appendix C.

3.3 Category-Extensible ObjectNav Task Construction

To support human-target search-and-rescue tasks, we need to introduce humanoid targets that are not originally present in existing navigation scenes and construct corresponding navigation episodes. We therefore propose *category-extensible ObjectNav*, where new semantic categories together with their embodied assets can be added to existing scenes while preserving the standard ObjectGoal observation, action, and

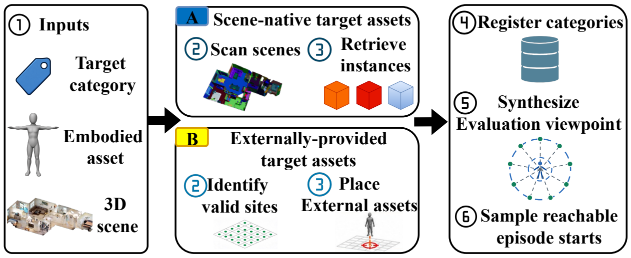

Figure 4: Category-extensible ObjectNav task construction pipeline.

evaluation interfaces. Fig. 4 illustrates the resulting construction pipeline, which converts both scene-native semantic objects and externally supplied embodied assets into simulator-compatible ObjectGoal episodes. For scene-native targets, the pipeline retrieves valid object instances from the scene inventory O; for externally supplied targets, it first identifies navigable and collision-free placement locations and instantiates the corresponding assets. Both target types are then registered under a unified semantic category space, followed by the generation of valid evaluation viewpoints and reachable episode starts. This allows scene-native and newly introduced categories to share the same episode-generation and navigation protocol. We instantiate this pipeline with humanoid assets to incorporate the *person* category and construct the PersonNav dataset, without requiring task-specific modifications to the downstream navigation framework. Detailed construction and evaluation procedures are provided in Appendix D.

4 TASK

We consider a *multi-agent search-and-rescue (SAR)* navigation task in dynamic indoor fire environments, where a team of mobile robots collaboratively explore an initially unknown scene, locate victims or other mission-critical objects, and avoid hazardous regions whose conditions evolve during deployment. Let $\mathcal{S} = \{s_1, \ldots, s_K\}$ the set of fire-affected indoor scenes and let $C = \{c_1, \ldots, c_m\}$ denote the set of target categories (e.g., *human*, *fire extinguisher*, *bathroom*). Unlike benchmarks limited to a fixed set of default goal vocabulary, our task allows the goal catalog to be extended with a novel category c_{new} and its corresponding embodied asset, while leaving the observation space, action space, planner interface, and evaluation protocol unchanged. We therefore refer to the task as *category-extensible* ObjectNav.

In each episode, N robots $\mathcal{R} = \{r_1, \ldots, r_N\}$ are deployed in a scene $s_i \in \mathcal{S}$ from a shared starting position $\mathbf{p}_i$ with different initial orientations (in our experimental setting, with their initial headings evenly distributed over 360°) to reflect realistic emergency deployment. All robots are assigned the

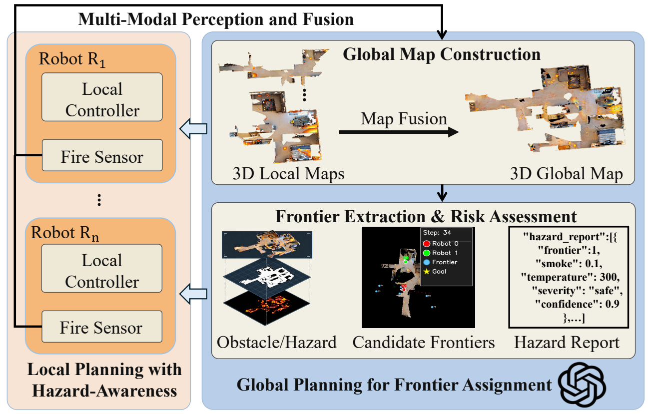

Figure 5: Overview of the unified FireNav framework.

same target category $c_i \in C$. The episode is defined as:

$$\mathcal{T}_i = \{N, s_i, c_i, \mathbf{p}_i\}. \tag{2}$$

The observation space O comprises four modalities: RGB, depth, thermal imaging, and mmWave radar from onboard sensors, providing complementary visual, geometric, and smoke-robust perception. The action space $\mathcal{A}$ consists of six discrete actions: {`move_forward`, `turn_left`, `turn_right`, `look_up`, `look_down`, `stop`}. The `move_forward` action advances the robot by 0.25 m, and each rotational action changes orientation by 30°. At each discrete time step t, robot r_i receives a multi-modal observation $o_t^i \in O$ and executes an action $a_t^i \in \mathcal{A}$ using its local observation history and information shared by the team.

An episode is considered successful if the team reaches a valid instance of the target category c_i, with the instance located within 0.2 m of an agent while ensuring that all agents remain within survivable hazard limits throughout the episode. The task therefore assesses the team's ability to jointly balance efficient exploration, semantic goal-directed navigation and risk-aware coordination in dynamic fire environments.

5 EVALUATION METHODOLOGY

As illustrated in Fig. 5, we propose a unified hazard-aware multi-agent navigation framework, FireNav, tailored for indoor fire SAR scenarios. Under this framework, each agent performs robust multi-modal perception by integrating RGB-D, thermal imaging, and mmWave radar observations to construct a local hazard-aware 3D representation. These local representations are aggregated into a shared global map and projected into aligned obstacle and hazard maps for frontier extraction and risk assessment. Based on this shared representation, navigation is organized into two hierarchical stages: global frontier assignment and local path execution. The global planner determines *where each robot should explore next* to enable efficient and risk-aware exploration in fire scenarios, whereas the local planner determines *how each robot reaches its assigned goal*, enabling real-time generation of smooth, collision-free, and risk-sensitive trajectories. This modular formulation decouples high-level exploration decisions from low-level motion execution and allows different planning strategies to operate on a common perception and mapping backbone.

5.1 Multi-modal Perception and Fusion

To maintain reliable situational awareness under such harsh conditions, our framework adopts a robust multi-modal perception pipeline that integrates RGB-D, thermal imaging, and mmWave radar signals. The goal is to (1) extract stable structural cues, (2) detect fire-related hazards, and (3) produce a unified obstacle-hazard map for downstream planning.

At each time step, the RGB, depth, and thermal images are first spatially and temporally aligned and processed by a model-agnostic fusion operator. An open-vocabulary detector and a class-agnostic segmentation model are then applied to the fused observation to identify semantic objects and fire-related regions. Each pixel u in the fused image, together with its depth $d(u)$, is back-projected to a 3D point in the camera frame:

$$\mathbf{x}_c(u) = d(u)\mathbf{K}^{-1}\tilde{u}, \tag{3}$$

where $\mathbf{K}$ is the camera intrinsic matrix and $\tilde{u}$ is the homogeneous pixel coordinate.

Thermal measurements are aligned with RGB pixels to estimate temperature $T(u)$, while smoke density $S(u)$ is inferred from RGB degradation and depth consistency. Each point is augmented with a hazard attribute vector:

$$\mathbf{h} = [T(u), S(u), \sigma(u)], \tag{4}$$

where $\sigma(u)$ encodes sensing uncertainty due to occlusion, smoke, or missing depth returns.

To obtain reliable geometric information under dense smoke, we leverage the smoke-penetrating capability of mmWave radar. Following prior LiDAR-supervised radar reconstruction approaches [15, 20], we train a radar reconstruction model offline using synchronized mmWave–LiDAR pairs, where LiDAR provides dense geometric supervision. At runtime, the trained model requires only mmWave measurements to reconstruct a dense, LiDAR-like point cloud.

The local hazard-aware 3D point cloud is constructed by fusing vision-based semantic and hazard points with LiDAR-like geometric points reconstructed from radar measurements. The local 3D point cloud maps from all robots are then transformed into a common coordinate frame and aggregated into a shared global hazard-aware 3D map. To mitigate spurious measurements caused by fire flickering, smoke dynamics, and thermal sensing artifacts, we apply DBSCAN [8] to the merged point cloud to remove sparse outliers.

5.2 Global Map Construction

To enable efficient risk-aware exploration in fire scenarios, a global 2D obstacle–hazard map is constructed from the 3D point cloud observations collected by the robots. The map provides a compact and physically grounded representation of both structural constraints and environmental hazards.

5.2.1 2D Obstacle-Hazard Map Construction. At the beginning of each episode, the global 3D map $\mathcal{M}$ is projected into a 2D grid-based representation centered at the robot's initial position. The resulting map consists of two aligned channels: an obstacle map $\mathbf{O}$ and a hazard map $\mathbf{H}$.

The obstacle map is constructed via top-down projection of non-floor 3D points, where grid cells exceeding a point-density threshold are marked as occupied. This produces a binary occupancy layout used to constrain navigation. The hazard map encodes fire-related environmental risks by aggregating hazard attributes from projected 3D points. Specifically, the hazard intensity of each grid cell is computed as a weighted combination of temperature, smoke density, and sensing uncertainty:

$$\mathbf{H}(x, y) = \sum_{x_g \in (x,y)} \left(w_T T + w_S S + w_\sigma \sigma \right), \tag{5}$$

where w_T, w_S, and w_σ balance the contributions of different hazard factors. Together, $\mathbf{O}$ and $\mathbf{H}$ form a compact obstacle–risk abstraction of the fire scene.

5.2.2 Frontier Extraction and Risk Assessment. Frontiers are defined as the boundaries between explored and unexplored regions after removing obstacle boundaries from $\mathbf{O}$. Frontier cells are clustered via connected-component analysis, and small noisy clusters are discarded. Each frontier is assigned a risk score by aggregating hazard intensities within its region. Let $\mathcal{F}_k$ denote the set of grid cells belonging to frontier k. The frontier-level risk is defined as:

$$R_k = \frac{1}{|\mathcal{F}_k|} \sum_{(x,y) \in \mathcal{F}_k} \mathbf{H}(x, y). \tag{6}$$

Frontiers are then categorized into qualitative risk levels (e.g., safe, moderate, and dangerous) using predefined thresholds. Based on the resulting frontier-level risk estimates, a structured hazard report is generated to summarize the hazard attributes of all candidate frontiers. This report provides a compact textual representation of environmental risk and is used as auxiliary input for VLM-based global planning.

5.3 Global Planning for Frontier Assignment

After frontier extraction and risk assessment, the global planning module determines a long-term exploration goal for each agent. Given the current robot states, global map, and frontier attributes, a global planner π^G produces a set of long-term goals:

$$\mathbf{g}_t^* = \pi^G(\mathcal{S}_t) = \{g_1^*, \dots, g_N^*\}, \tag{7}$$

where $\mathcal{S}_t$ summarizes the current planning state, including the obstacle map, hazard map, robot poses, and candidate frontiers. Frontier-based planners select a goal $f_i^* \in \mathcal{F}_t$ for each agent r_i, while alternative strategies may select from other feasible map locations. We formulate global goal assignment through a common planner interface that supports stochastic sampling, cooperative greedy exploration [29], cost–utility optimization [7], and VLM-based planning [14, 25, 39]. All strategies operate on the same map representation, frontier information, and robot states, enabling controlled comparison of different exploration and coordination strategies. Hazard information is exposed through the same interface and can be incorporated by individual planners as a soft preference when selecting long-term goals. Detailed implementations of the evaluated global planners are provided in Appendix F.1.

5.4 Local Path Planning

Given an assigned long-term goal, the local planning module determines a feasible path from the current robot pose to the target location, jointly considering geometric feasibility and environmental risk. We consider two representative categories of local navigation methods: *field/search-based planners*, including FMM [24] and A* [5], which explicitly reason over spatial maps and traversal costs, and a *reinforcement-learning-based planner*, PointNav DD-PPO [33], which directly predicts navigation actions from sensory observations. All methods are evaluated through a common local-planning interface under identical global assignments and map representations. Detailed baseline implementations are provided in Appendix F.2.

6 EXPERIMENTAL EVALUATION

To systematically evaluate FireNav, we organize our experiments around the following five research questions (RQs): **RQ1 – Planning Effectiveness.** How do different global and local planning strategies perform under a common navigation framework across diverse environments? **RQ2 – Adaptation to Dynamic Fire.** How effectively does FireNav maintain navigation performance and safety under dynamic fire conditions compared with hazard-free environments? **RQ3 – Component Effectiveness.** To what extent do multimodal perception and hazard-aware planning mitigate fire-induced performance degradation? **RQ4 – Robustness and Generalization.** How robust are the observed performance trends across different scene scales, fire types and intensities, and VLM backbones? **RQ5 – Physical Feasibility.** Can FireNav transfer to physical multi-robot platforms while maintaining practical online performance?

6.1 Experimental Setup

Scene Dataset. We conduct all simulation experiments on the FireNav benchmark over 36 HM3DSem validation scenes. The benchmark comprises two task subsets. For scene-native object search, we retain the standard Habitat ObjectNav episode definitions and evaluate them under the dynamic fire conditions generated by FireWorld. This subset contains 1,000 episodes spanning six representative categories: *chair*, *sofa*, *plant*, *bed*, *toilet*, and *TV*. To additionally evaluate search-and-rescue scenarios involving human targets, we construct a PersonNav subset using our category-extensible task pipeline, yielding 623 episodes over the same scenes while preserving the standard navigation interface and evaluation protocol.
Setting. We instantiate the benchmark in the Habitat simulator with multiple mobile robots using Spot and Fetch embodiments. For consistent navigation and collision evaluation across embodiments, each robot is modeled with a height of 1.41 m and a base radius of 0.17 m. Robots start from the same location with different initial orientations and share a common map. Each robot receives 480×640 RGB-D observations, odometry information, simulated thermal and mmWave radar measurements, and a semantic goal category. We use YOLOv8 [6] for open-vocabulary object detection and Mobile SAM [11] for class-agnostic segmentation. Multi-robot observations are projected onto a shared 24×24 m exploration map with 0.05 m resolution. For the VLM-based global planner, we use GPT-4o through the OpenAI API. Unless otherwise specified, all settings within each research objective are evaluated on 200 matched episodes, with identical perception, mapping, and episode configurations.

6.2 Evaluation Metrics

Our evaluation considers three complementary aspects of navigation: task performance, navigation efficiency, and safety. We report six metrics in total: Success Rate (SR), Distance to Goal (DTG), Success weighted by Path Length (SPL), Number of Steps (NS), Average Hazard Exposure (AHE), and Severe Hazard Exposure Rate (SHER).
Task Performance. SR measures the percentage of episodes in which at least one robot successfully reaches a valid goal location. DTG measures the minimum geodesic distance between any robot and the target at episode termination, providing a measure of progress even when an episode is unsuccessful. Higher SR and lower DTG indicate better task performance.
Navigation Efficiency. NS measures the steps taken before episode termination. SPL jointly considers task success and

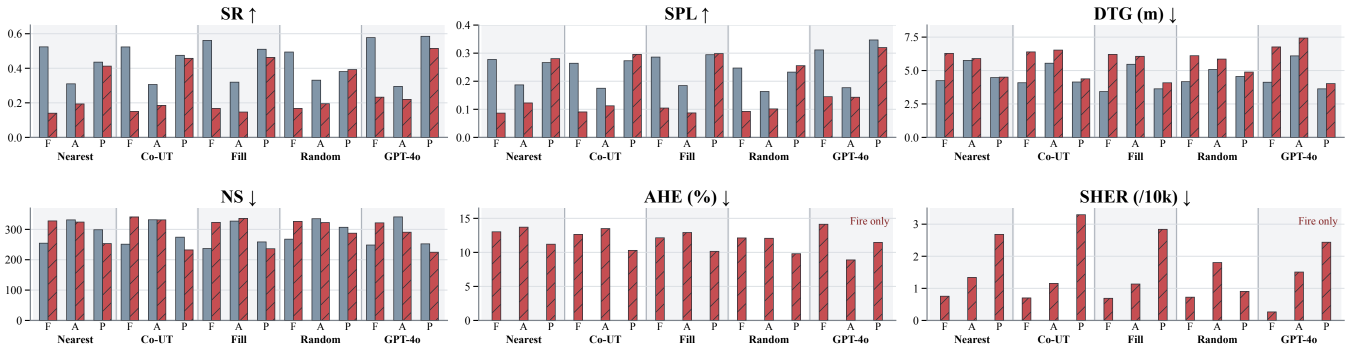

Figure 6: Conventional navigation performance under hazard-free and dynamic-fire conditions, averaged across PersonNav and ObjectNav. Gray solid and red hatched bars denote hazard-free and dynamic-fire conditions, respectively. F, A, and P denote FMM, A*, and PointNav; AHE and SHER are reported only under dynamic fire.

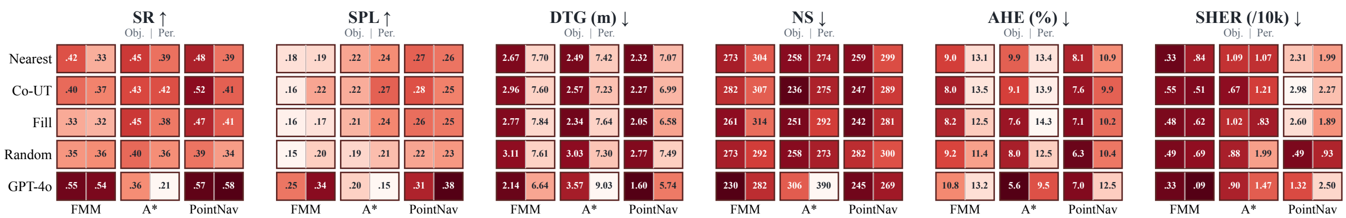

Figure 7: Performance heatmap of FireNav under dynamic fire conditions. Each cell reports ObjectNav (left) and PersonNav (right), with darker shades indicating better results.

path efficiency by comparing the executed trajectory with the shortest path to a valid goal location. Lower NS and higher SPL indicate more efficient navigation.

Navigation Safety. AHE measures the average hazard intensity experienced across all agent-steps, capturing the overall level of risk encountered during navigation. SHER measures the fraction of agent-steps for which the experienced hazard exceeds a predefined severe-hazard threshold, capturing how frequently robots enter high-risk regions. Lower AHE and SHER indicate safer navigation.

6.3 Overall Performance

To answer RQ1 and RQ2, we evaluate global–local planner baselines on ObjectNav and PersonNav under both hazard-free and dynamic-fire conditions (Table 4), and assess the full FireNav framework under dynamic fire (Table 5). We analyze these results from three perspectives: planning effectiveness, the impact of dynamic fire hazards, and the overall effectiveness of FireNav.

Planning Effectiveness. As shown in Fig. 6, global and local planners exhibit distinct yet complementary behavior across tasks and environmental conditions. Averaged across local planners, GPT-4o achieves the highest SR on both ObjectNav and PersonNav under both hazard-free and dynamic-fire conditions, highlighting the benefit of semantic reasoning for long-horizon exploration. The choice of local planner is likewise important, but its effectiveness depends on the environment: under hazard-free conditions, FMM achieves the highest average SR on ObjectNav (0.566) and PersonNav (0.506), whereas PointNav performs best under dynamic fire (0.410 and 0.486, respectively).

No single global–local combination dominates all metrics. Under hazard-free conditions, GPT-4o with PointNav achieves the highest PersonNav SR (0.590) and SPL (0.380), while Fill with FMM attains the highest ObjectNav SR (0.597) with the fewest steps (197.1). Across conditions, PersonNav generally exhibits lower success and longer trajectories than ObjectNav, indicating a more challenging search setting. Overall, these results highlight the complementary roles of global coordination and local execution rather than suggesting a universally optimal planner.

Impact of Dynamic Fire Hazards. Fig. 6 compares identical planner configurations under hazard-free and dynamic-fire conditions. Averaged across both tasks and all planner combinations, dynamic fire reduces SR and SPL by 39.1% and 31.2%, respectively, while increasing hazard exposure. This degradation is consistent across planners, highlighting the fundamental challenge posed by evolving fire hazards.

Performance of FireNav. Fig. 7 summarizes the performance of the FireNav framework. FireNav remains effective

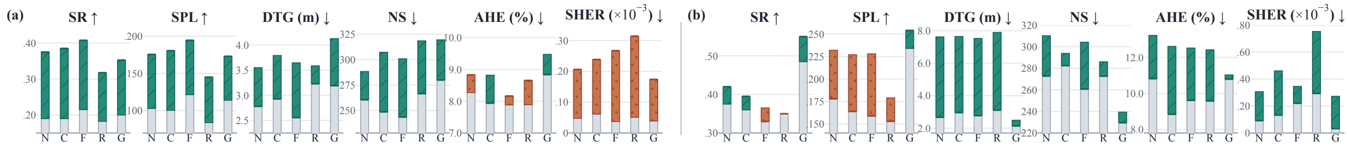

Figure 8: Ablation study of multi-modal perception and fusion (a) and risk-aware planning (b). Green hatched (orange dotted) segments indicate improvements (regressions). N/C/F/R/G denote Nearest/Co-UT/Fill/Random/GPT-4o.

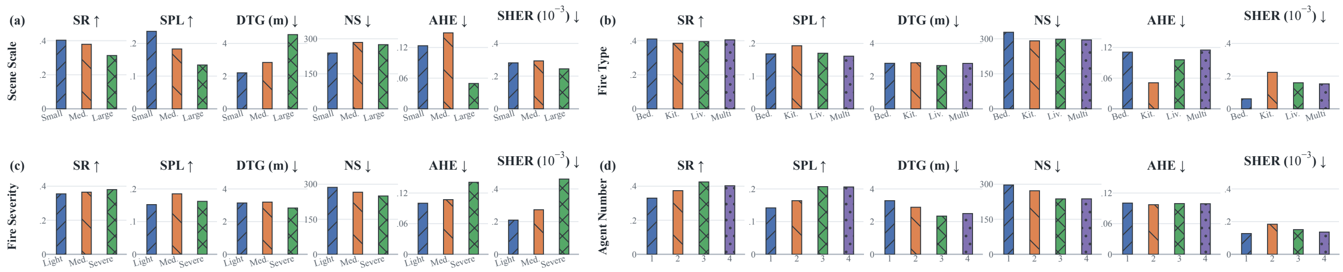

Figure 9: Sensitivity of navigation performance to scene scale, fire type, fire severity, and team size: (a) scene scale, (b) fire type, (c) fire severity, and (d) number of agents.

across both ObjectNav and PersonNav, improving average SR and SPL by 53.0% and 35.7%, respectively, over conventional navigation under the same fire conditions, while reducing AHE and SHER by 15.0% and 20.5%. GPT-4o with PointNav achieves the highest SR (0.570 and 0.585) and SPL (0.312 and 0.377) on ObjectNav and PersonNav, respectively, whereas GPT-4o with FMM yields the lowest SHER on both tasks. Overall, these results show that FireNav can effectively operate under evolving fire conditions.

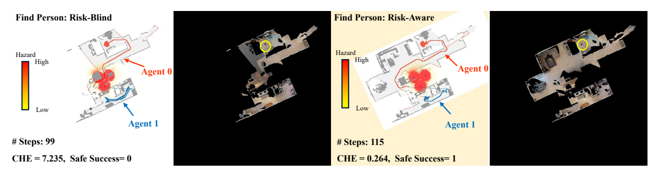

Figure 10: Qualitative comparison of risk-blind (left) and risk-aware (right) cooperative navigation under PersonNav in the same dynamic fire scenario.

6.4 Ablation Study

To answer RQ3, we evaluate the contributions of two key components of FireNav through ablation studies. Given FMM's consistently strong performance across global planners, we fix it as the local planner to isolate the effects of perception enhancement and risk-aware planning. Each setting is evaluated on 200 matched episodes.

Multi-modal Perception and Fusion. We first evaluate the effect of multi-modal perception and fusion by comparing conventional RGB-D sensing with our multi-modal perception stack. All other components, including mapping pipeline, and planner configurations are kept identical. As shown in Fig. 8 (a), enhanced perception consistently improves task completion and navigation efficiency across all five global planners, achieving SRs of 0.318–0.408 and SPLs of 0.145–0.195, while reducing DTG to 2.557–3.231 m and NS to 243.7–280.1 steps. Notably, these gains also extend to planners that explicitly model environmental hazards, indicating that effective risk-aware decision making still relies on reliable upstream perception. The results highlight the complementary roles of robust perception and high-level coordination in enabling efficient and safe cooperative navigation.

Risk Awareness. We next evaluate the effect of hazard-aware planning by enabling risk awareness in both global frontier assignment and local path execution. As shown in Fig. 8 (b), risk-aware planning substantially reduces hazard exposure across all global planners, with average AHE and SHER decreasing by 16.1% and 62.0%, respectively. At the same time, DTG decreases by 59.7% and SR improves from 0.359 to 0.409. The moderate decrease in SPL, from 0.209 to 0.182, reflecting a safety–efficiency trade-off as agents take safer detours. Notably, GPT-4o with risk awareness achieves the highest SR (0.550) and SPL (0.254) and the lowest SHER (0.0327), demonstrating that hazard information can be effectively incorporated without sacrificing task performance. Overall, these results highlight the importance of explicitly balancing navigation progress and hazard exposure in dynamic fire environments. Fig. 10 further illustrates how hazard-aware planning redirects agents toward safer routes.

6.5 Microbenchmark Evaluation

To answer RQ4, we analyze the sensitivity of navigation performance to scene scale, fire type, fire severity, and team size, as summarized in Fig. 9. In our microbenchmark experiments, we use Co-UT and FMM as the global and local planners, respectively, and vary one factor at a time while holding all others constant.

Impact of Scene Scale. We divide the 36 scenes into three equal-sized groups using tertiles of their largest connected navigable area: small (20.6–63.5 m^2), medium (64.1–101.0 m^2), and large (103.2–275.0 m^2). For each group, we evaluate 200 ObjectNav episodes with medium-severity multi-origin fires. As shown in Fig. 9(a), SR and SPL decrease from 0.403 to 0.313 and from 0.237 to 0.134 as scene size increases, with large scenes requiring 15.0% more steps. The lower AHE and SHER suggest that sparser layouts offer greater flexibility for safe rerouting and reduced hazard exposure.

Impact of Fire Type. Fig. 9(b) compares four fire types at medium severity. SR remains similar across settings (0.385–0.410), while multi-origin fires yield the lowest SPL (0.161) and highest AHE (0.115), suggesting that more spatially complex fire configurations can increase navigation cost and hazard exposure. Overall, fire type has limited impact on task completion, but may notable influence navigation efficiency and safety.

Impact of Fire Severity. Fig. 9(c) compares light, medium, and severe multi-origin fires across the same set of scenes. SR remains relatively stable (0.355–0.380), whereas AHE increases from 0.100 to 0.141 and SHER from 0.209×10^{-3} to 0.460×10^{-3}. These results indicate that higher fire severity primarily increases the navigation risks, while task performance and navigation efficiency remain relatively stable.

Impact of Agent Number. As shown in Fig. 9(d), increasing the team size generally improves navigation performance and efficiency through parallel exploration. Averaged across global planners, SR increases from 33.00% with one agent to 42.50% with three agents, while SPL rises from 0.142 to 0.207 and NS decreases from 296.8 to 237.3. However, adding a fourth agent yields little further benefit, with SR decreasing slightly to 40.25% and SPL and NS remaining nearly unchanged. These results demonstrate clear multi-agent gains but also diminishing returns as team size increases.

6.6 Physical Experiments

We deploy our framework on two Unitree Go2 quadruped robots, each equipped with an Intel RealSense D455 RGB-D camera and a Livox Mid-360 LiDAR, as shown in Fig. 17. The real world navigation task includes searching for a bed and a person in a fire environment.

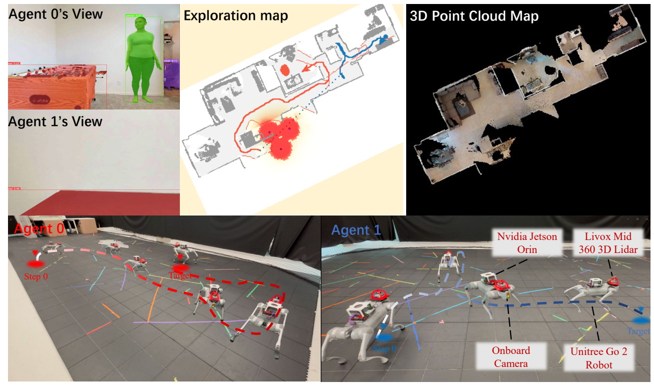

Figure 11: Real-world multi-robot navigation to find a person.

Each robot uses the onboard Livox Mid 360 LiDAR and its integrated IMU for localization. FAST LIO2 [36] is employed to estimate the robot pose, which is continuously synchronized with the pose representation used by the navigation framework. Based on the current map and robot poses, the framework generates navigation actions together with their corresponding motion trajectories in the simulation environment. These trajectories are then sent to the real robots, which track them using the LiDAR based pose estimates. In this way, high level perception and planning remain unchanged from the simulation setup, while trajectory tracking and low level motion execution are performed on the physical robots. Fig. 11 shows a representative real-world deployment in which two quadruped robots cooperatively search for a person in a fire environment.

All perception and high-level planning modules run on a workstation with an NVIDIA RTX 4060 Ti GPU, while localization, trajectory tracking, and low-level control run onboard at approximately 10 Hz. Table 3 reports the mean and standard deviation of runtime over 100 executions of each module. The navigation loop operates at approximately 4 Hz on our platform. VLM-based global planning requires an additional 2.1 s on average, but is invoked only when global replanning is required; local navigation continues at its normal control frequency between global planning updates. These results demonstrate that the framework can support closed-loop real-world multi-robot navigation with practical runtime overhead.

7 DISCUSSION & LIMITATION

Backbone for dynamic embodied tasks. The design of FireNav is not tied to a specific simulator or navigation dataset. The same pipeline can be adapted to other simulators and datasets with compatible scene representations. This makes FireNav a potential backbone for studying embodied agents in complex dynamic environments beyond fire navigation. In particular, the evolving hazard state can be coupled with other embodied tasks, such as manipulation and interactive exploration, where agents must reason about both task objectives and changing environmental risks.

Fidelity and scalability trade-off. FireNav targets a different operating point from engineering-grade fire simulators. For embodied navigation, the objective is not to reproduce every fluid-dynamic detail, but to preserve the spatial and temporal hazard structure relevant to perception and decision making while enabling scalable and reproducible evaluation. Decoupling lightweight hazard propagation from online interaction allows FireWorld to retain dynamic fire evolution without placing expensive physical simulation in the robot-control loop.

Safety and efficiency trade-off. Navigation in dynamic fire environments inherently requires balancing safety and efficiency. Avoiding hazardous regions can reduce fire exposure but may introduce longer detours or delay exploration, while shortest-path decisions may incur substantially higher risk. FireNav therefore incorporates risk as a soft planning cost, allowing global and local planners to balance navigation progress against exposure. An important direction is to develop adaptive planning strategies that dynamically adjust this balance according to hazard severity, mission urgency, and the remaining navigation budget.

Toward predictive hazard-aware navigation. The current framework primarily reacts to the estimated hazard state at planning time. However, in an evolving fire, a route that is currently safe may become hazardous before the robot reaches it. FireWorld's time-varying hazard representation provides a natural basis for future predictive planners that jointly reason about travel time and future hazard evolution.

Limitations. FireNav employs a lightweight physics model rather than full CFD and does not capture all fire phenomena, such as detailed ventilation effects or structural changes. Its sensor models reproduce modality-level degradation trends but do not fully model hardware-specific effects. Hazard evolution is currently precomputed, preventing robot actions from affecting the fire world. Our multi-agent evaluation also assumes reliable shared communication, while the physical validation is limited to two robots and does not reproduce the complete simulated sensor suite. Future work will incorporate calibrated sensor models, communication-constrained coordination, bidirectional environment interaction, and larger-scale real-world evaluation.

8 RELATED WORK

Benchmarks for Embodied Navigation. Embodied navigation benchmarks such as Habitat and HM3D provide realistic 3D environments and standardized interfaces for tasks including PointNav and ObjectNav [22, 23]. Recent benchmarks further extend embodied navigation toward open-vocabulary and lifelong goals [9, 38], interactive tasks [35], disaster response [41], and multi-agent navigation [2, 13, 31]. However, existing benchmarks do not jointly support physics-grounded evolving hazards, cooperative multi-agent navigation, and extensible goals; FireNav unifies these capabilities within a common benchmark.

Multi-agent Cooperative Navigation. Prior multi-agent navigation studies have investigated coordinated exploration and communication-aware collaboration [1, 29], while learning-based methods further improve cooperative visual exploration and task allocation [40]. Recent approaches leverage VLMs for semantic reasoning and coordinated goal assignment [25, 39], with emerging efforts extending multi-agent navigation toward fire-disaster response and dynamic threats [13, 14]. However, cooperative navigation under evolving hazards remains underexplored; FireNav addresses this gap with unified hazard-aware multi-agent perception and planning.

Search and Rescue (SAR). Robotic SAR has been widely studied across aerial, ground, and marine platforms [3, 16], with multi-robot systems enabling cooperative search and situational awareness [21]. Prior simulation studies range from abstract disaster-response platforms [28] and handcrafted robotic scenarios [26] to recent dynamic embodied environments [41]. FireNav complements these efforts with scalable cooperative SAR under physics-grounded evolving indoor fire, jointly modeling hazard dynamics, degraded multimodal sensing, and multi-agent navigation.

9 CONCLUSION

We present FireNav, the first physics-based and scalable benchmark for cooperative multi-agent navigation in dynamic indoor fire environments. Its underlying infrastructure, FireWorld, transforms static 3D scenes into controllable and reproducible dynamic fire worlds with evolving hazards, fire-aware multimodal sensing, and category-extensible navigation goals. Building on this infrastructure, FireNav provides a unified framework for multimodal perception, cooperative planning, and risk-aware navigation. Extensive evaluations across ObjectNav and PersonNav, diverse planners, scenes, fire conditions and team sizes demonstrate that dynamic hazards substantially challenge conventional navigation, while FireNav adapts effectively to such environments through multimodal perception and hazard-aware planning, enhancing both robustness and safety. We hope FireNav serves as a reusable foundation for developing and evaluating embodied intelligence in complex, dynamic, and hazardous environments beyond fire navigation.

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A Dynamic Fire Scenario Construction

A.1 Trial and Error

Our initial approach to constructing dynamic fire environments was to directly modify the underlying scene assets. Specifically, we edited the original `.glb` scene files in Blender and augmented them with fire- and smoke-related visual effects. Semi-transparent textured billboards were used to approximate smoke, while animation was applied to produce time-varying smoke motion. This approach can generate visually realistic dynamic fire scenes in a general-purpose rendering environment and, at first glance, provides a straightforward way to convert an existing indoor scene into a fire-affected environment.

We subsequently attempted to import these modified assets into both Habitat and Isaac Sim. In practice, however, this asset-level approach proved difficult to integrate reliably with embodied-navigation simulators. Their rendering pipelines are primarily designed for efficient robotic perception, interaction, and physics simulation, rather than for faithfully reproducing arbitrary animated transparency and complex time-varying visual effects authored in external graphics tools. As a result, transparency, animated smoke layers, and related effects were not consistently preserved after import, making direct patching of the scene `.glb` files unsuitable for our benchmark construction pipeline.

More importantly, even if such visual effects could be reproduced faithfully, asset-level animation would provide only a visually plausible approximation of fire. The appearance of flames and smoke would remain largely decoupled from physically meaningful quantities such as temperature, fuel availability, smoke density, and spatial hazard propagation. This limitation motivated us to move away from purely asset-based scene editing and instead represent fire as an explicit time-varying environmental state. In FireWorld, fire, smoke, and temperature are therefore modeled as spatial fields and subsequently rendered into the simulator observations. This design separates hazard evolution from scene graphics, improves reproducibility across navigation policies, and provides a physically grounded state that can be directly used for sensing, risk assessment, and navigation.

A.2 Scene Inventory Construction

The scene inventory provides the spatially grounded world model used by subsequent fire-scenario generation. Unlike standard ObjectNav annotations, which are primarily concerned with navigation targets, we recover *all* semantic instances in each HM3D scene. Each recovered instance is associated with its semantic identity, spatial extent, floor assignment, and fire-related physical properties. Structural elements are additionally identified to constrain hazard propagation. As a result, the inventory provides a unified representation for semantic ignition selection, physical property grounding, and geometry-aware fire propagation.

Listing 1: Abridged example of the generated scene inventory.

```
{
  "scene_id": "4ok3usBNeis",
  "voxel_m": 0.1,
  "instances": [
    {
      "instance_id": 8,
      "category": "chair",
      "aabb_min": [2.1079, -0.5986, -0.6238],
      "aabb_max": [2.8500, 0.4104, 0.0778],
      "structural": false,
      "is_goal": true,
      "flammability": 0.55,
      "smoke_yield": 0.50
    },
    {
      "instance_id": 25,
      "category": "floor",
      "aabb_min": [-0.5948, -0.9355, -0.8143],
      "aabb_max": [3.4623, -0.4186, 6.2657],
      "structural": true,
      "is_goal": false,
      "flammability": 0.0,
      "smoke_yield": 0.0
    },
    ...
  ],
  "build_summary": {
    "n_instances_in_txt": 513,
    "n_instances_recovered": 508,
    "n_structural_instances": 136,
    "n_flammable_instances": 366
  }
}
```


Figure 12: Representative instantiations of the four semantic fire templates in FireWorld: Kitchen Grease Fire, Bedroom Textile Fire, Living-room Electrical Fire, and Multi-origin Fire.

Semantic instance recovery. For each HM3D scene, we jointly parse the corresponding `*.semantic.glb` and `*.semantic.txt` files to recover all semantic instances and their geometry. Instance identities are encoded by per-vertex colors in the semantic mesh. Specifically, the stored `uint16` linear RGB values are normalized, converted to sRGB following IEC 61966-2-1, quantized to 24-bit colors, and matched to the instance identifiers in `*.semantic.txt`. Since a single GLB mesh primitive may contain geometry from multiple semantic instances, we assign instance labels at the face level and aggregate faces sharing the same identifier. This procedure recovers the complete instance-level scene content rather than only the predefined ObjectNav target categories.
Spatial grounding. The recovered geometry is transformed into Habitat world coordinates. For each instance, we compute its axis-aligned bounding box (AABB) and centroid and associate it with a floor according to its vertical extent. These quantities provide the spatial support required for ignition placement and object-level hazard reasoning.
Physical and structural grounding. Each semantic category is mapped to fire-related properties, including *flammability* and *smoke yield*, through a category-level lookup table. Structural elements, such as floors, walls, ceilings, windows, and other solid boundaries, are separately marked and rasterized into a voxelized structural mask at a resolution of 0.10. This mask is used by the propagation solver to prevent heat and smoke transport through solid scene geometry.

Inventory representation. The resulting inventory is serialized into a machine-readable JSON file. Each recovered instance records its semantic identity, spatial extent, structural role, navigation-target association, and fire-related physical properties. Listing 1 shows an abridged example from one HM3D scene.

A.3 Fire Semantic Templates

FireWorld uses semantic fire templates to generate diverse yet reproducible fire scenarios across indoor scenes. We consider four representative templates: *Kitchen Grease Fire*, *Bedroom Textile Fire*, *Living-room Electrical Fire*, and *Multi-origin Fire*. The first three represent localized fires associated with different room and object contexts, whereas the Multi-origin template introduces multiple ignition sources distributed across the scene. Representative examples are shown in Fig. 12. The examples illustrate how the template definition affects the spatial distribution of flame and smoke while preserving the geometry and semantics of the underlying indoor environment.
Candidate construction. For a template c, we first construct a candidate set $\mathcal{E}_c \subseteq \mathcal{O}$ from the scene inventory. Each template defines a set of preferred semantic categories together with fallback categories. Preferred categories encourage semantically consistent ignition placement, while the fallback set ensures that a valid fire scenario can still be generated when the preferred objects are absent from a

particular scene. Structural and non-combustible objects are excluded from the candidate set.

Ignition objects are not selected uniformly at random. Instead, we borrow a physics-inspired prior motivated by two observations on real fire scenarios: more combustible objects are more likely to ignite and sustain combustion, while different fire scenarios are typically associated with distinct object categories and room contexts. Therefore, this design can generate semantically coherent and physically plausible benchmark scenarios.

Primary ignition sampling. Candidate objects are sampled according to their material-dependent flammability rather than uniformly. Let w_i denote the flammability assigned to candidate object i. The probability of selecting i as an initial ignition source is

$$P(i \mid c) = \frac{w_i}{\sum_{j \in \mathcal{E}_c} w_j}, \qquad i \in \mathcal{E}_c. \tag{8}$$

Thus, objects that are more likely to sustain combustion receive a higher ignition probability while the semantic template constrains the overall scenario context.

Secondary ignition sampling. For templates requiring additional ignition sources, we additionally consider spatial proximity to an already selected anchor source. Given an anchor i, a secondary source j is sampled according to

$$P(j \mid i, c) \propto w_j \, \psi\left(\left\|\mathbf{p}_j - \mathbf{p}_i\right\|_2\right) \mathbf{1}\left[\ell_j = \ell_i\right], \tag{9}$$

where $\mathbf{p}_j$ and ℓ_j denote the position and floor index of candidate j, respectively, and $\psi(\cdot)$ is a distance-weighting function. This formulation favors nearby combustible objects on the same floor, producing spatially coherent fire clusters rather than independent random ignition points. Depending on the template, selected sources can be activated immediately or after a template-defined delay, supporting both simultaneous and staged ignition.

Propagation Configuration and Plan Representation. Three intensity presets, $q \in \{light, medium, severe\}$ control the primary-source temperature, per-ignition fuel mass, and simulated duration across all four templates. It further specifies the downstream smoke-generation and propagation parameters, including ignition thresholds, flame-spread and ceiling-jet rates, buoyancy, thermal diffusion, decay, and ambient cooling. Together with the scenario template, these values instantiate $\boldsymbol{\theta}_{\text{prop}}$ and thereby govern the evolution of the flame, smoke, and temperature fields. The resulting fire plan is serialized as a JSON file that records the scenario type, intensity, seed, propagation parameters $\boldsymbol{\theta}_{\text{prop}}$, and the source-level attributes of each ignition in $\mathcal{I}$, including its world position, radius, temperature, ignition time, and fuel mass. The plan is both machine-readable and human-editable, allowing users to add ignition sources or modify their timing and strength without changing the simulation pipeline.

Figure 13: Temporal evolution of representative fire scenarios. Each row shows the early, middle, and late stages of one semantic template.

Automatically generated and edited plans are processed by the same propagation model, enabling controlled scenario variations under a consistent simulation procedure.

A.4 Temporal Evolution of Fire Scenarios

Fig. 13 further illustrates the temporal evolution of the same four template types at representative early, middle, and late stages. Unlike static hazard placement, fire in FireWorld evolves as a time-varying environmental process. As the simulation progresses, active ignition sources generate flame, heat, and smoke, which subsequently propagate through spatially connected regions according to the physics-inspired voxel model described in Sec. 3.1.3. The sequences highlight both temporal and scenario-level diversity. Localized templates exhibit progressive growth around their semantic ignition regions, while the multi-origin configuration produces a broader and more spatially heterogeneous hazard distribution due to multiple simultaneously or sequentially active sources. These time-varying fields are subsequently baked into the fire-scene timeline and replayed during navigation, allowing agents to encounter different hazard conditions as an episode progresses.

Table 1: Runtime efficiency of FireWorld. Online latency is measured for joint observations from two agents at 640×480.

Module	Runtime	Throughput
Offline timeline baking	150.2 s / timeline	3.98× real time
Navigation-fast	74.85 ms / observation	13.36 FPS
Full-quality	225.73 ms / observation	4.43 FPS

A.5 Time Overhead

Baking Time of Fire Timelines. Fire propagation is precomputed offline and therefore does not introduce latency into the online navigation loop. Across 36 fire timelines, FireWorld requires 1.502 hours in total, corresponding to 150.2 s per timeline on average. More importantly, the baking process achieves 3.98× real-time simulation throughput, enabling dynamic fire scenarios to be generated efficiently before evaluation.

Online Rendering Time. We measure the rendering latency using two agents with 640×480 observations, excluding perception, mapping, and planning. The `navigation_fast` mode used in our benchmark requires only 74.85 ms per joint observation, corresponding to 13.36 FPS. Thus, although FireWorld introduces additional rendering computation, it remains sufficiently fast for online embodied navigation. The full-quality mode achieves 4.43 FPS and is mainly used for high-fidelity visualization.

B Fire Propagation Details

We provide additional details of the physics-inspired voxel propagation model described in Sec. 3.1.3. The environment is discretized into a regular voxel grid with resolution 0.15, and the propagation is simulated with a fixed timestep of $\Delta t = 0.5$.

At voxel center $\mathbf{p}$ and simulation step n, the local fire state is represented as

$$\mathbf{x}^n(\mathbf{p}) = (F^n(\mathbf{p}), T^n(\mathbf{p}), L^n(\mathbf{p}), S^n(\mathbf{p})), \tag{10}$$

where $F \in [0,1]$ denotes the remaining fuel fraction, T the temperature, $L \in [0,1]$ the flame intensity, and $S \in [0,1]$ the smoke density. Structural voxels are excluded from propagation so that walls and other solid geometry constrain transport.

Heat and smoke transport. Heat first diffuses between neighboring free-space voxels. The supra-ambient heat and smoke are then transported upward to approximate buoyant plume motion:

$$\widetilde{T} = T^n + \alpha_T \Delta t\, \nabla_h^2 T^n, \tag{11}$$

$$q_m^{\mathrm{tr}}(\mathbf{p}) = q_m\left(\mathbf{p} - v_m \Delta t\, \mathbf{e}_y\right), \qquad m \in \{T, S\}, \tag{12}$$

where $q_T = \max(\widetilde{T} - T_{\mathrm{ambient}}, 0)$, $q_S = S^n$, and v_m is the corresponding buoyant rise speed. The transported fields are reconstructed as $T^{\mathrm{tr}} = T_{\mathrm{ambient}} + q_T^{\mathrm{tr}}$ and $S^{\mathrm{tr}} = q_S^{\mathrm{tr}}$. Near ceilings, the rising heat and smoke are additionally redistributed laterally in the horizontal plane to approximate ceiling-induced plume spreading.

Combustion and local flame spread. A voxel is considered burning when sufficient fuel remains and its temperature exceeds the ignition threshold, i.e., $I_{\mathrm{burn}} = \mathbf{1}[F^n > \phi_f]\mathbf{1}[T^{\mathrm{tr}} > T_{\mathrm{ig}}]$. The combustion update is

$$\begin{aligned}
F^* &= \mathrm{clip}_{[0,1]}\left(F^n - r_F \Delta t\, I_{\mathrm{burn}}\right),\\
T^* &= T^{\mathrm{tr}} + r_T \Delta t\, I_{\mathrm{burn}},\\
L^* &= \mathrm{clip}_{[0,1]}\left(L^n + r_L \Delta t\, I_{\mathrm{burn}} + \kappa_{\mathrm{spread}} \Delta t\, \mathbf{1}[F^n > 0] \nabla_h^2 L^n\right),\\
S^* &= \mathrm{clip}_{[0,1]}\left(S^{\mathrm{tr}} + r_S \Delta t\, I_{\mathrm{burn}}\right),
\end{aligned} \tag{13}$$

where r_F controls fuel consumption, r_T, r_L, r_S control the generation of heat, flame, and smoke, and κ_{spread} controls local flame spread. The fuel mask prevents flame activity from spreading persistently through non-combustible regions.

Dissipation and source forcing. After combustion, flame and smoke gradually dissipate, while temperature relaxes toward ambient conditions:

$$\begin{aligned}
L^{\mathrm{d}} &= L^* e^{-\lambda_L \Delta t}, \qquad S^{\mathrm{d}} = S^* e^{-\lambda_S \Delta t},\\
T^{\mathrm{d}} &= T_{\mathrm{ambient}} + (T^* - T_{\mathrm{ambient}})\, e^{-\lambda_T \Delta t}.
\end{aligned} \tag{14}$$

Finally, ignition sources specified by the generated fire plan reapply localized heat, flame, and smoke forcing while their prescribed activation intervals remain active. The resulting state forms $\mathbf{x}^{n+1}$, and the procedure is repeated to generate the time-varying hazard fields used by the rendering pipeline.

C Fire Sensor Data Synthesis

C.1 Degraded Sensing

Fire dynamics affect not only where a robot can safely move but also what it can reliably perceive. Dense smoke and elevated temperatures can substantially alter the performance of sensing modalities commonly used for robotic navigation. Experimental studies have evaluated visible and infrared cameras, LiDAR, depth cameras, sonar, and radar under fire-smoke conditions, demonstrating that sensing reliability varies considerably across modalities and environmental conditions [15, 20, 27, 30]. Thermal imaging and radar can provide complementary information when visible-spectrum observations become unreliable; prior firefighting-robot systems, for example, have combined thermal infrared vision with FMCW radar to recover scene structure through dense smoke [10]. These observations motivate a fire-navigation benchmark in which the evolving physical hazard state is explicitly coupled to the agent's observation process, rather

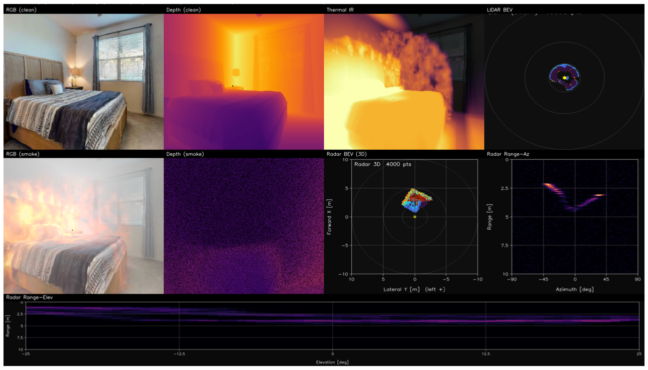

Figure 14: Representative multi-modal observations for a bed-search scenario under dynamic fire.

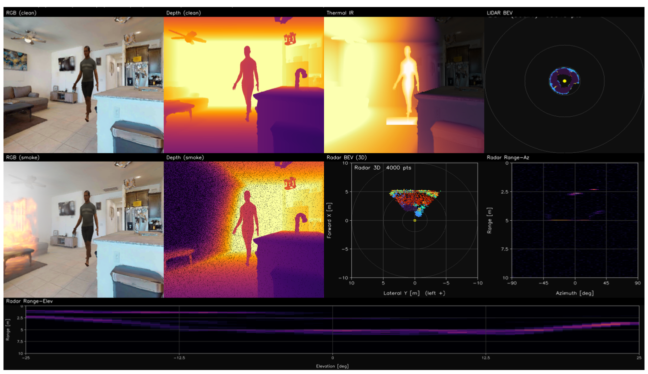

Figure 15: Representative multi-modal observations for a human-search scenario under dynamic fire.

than treating fire solely as a visual effect or a static cost region.

C.2 Representative Sensor Observations

Figs. 14 and 15 show representative multi-modal observations generated by the FireWorld fire-sensor suite for object- and human-search scenarios, respectively. RGB, depth, thermal, and radar-based observations are synthesized from the same synchronized fire state, providing complementary visual, geometric, and thermal cues under fire-induced sensing degradation. These examples illustrate how FireWorld couples the evolving hazard field with the agent's observation process while preserving cross-modal spatial and temporal consistency.

D PersonNav Dataset Construction

Person target placement. We instantiate the category-extensible pipeline using a static humanoid asset to represent the *person* category. Candidate placement locations are filtered according to navigability, collision, and scene-consistency constraints. Unlike scene-native ObjectNav targets, person locations are not derived from the original semantic annotations, reducing reliance on conventional object–room and object–object co-occurrence priors.

Evaluation viewpoint generation. Candidate evaluation viewpoints are sampled on a 5-cm Cartesian lattice within 0.20–1.10 m of the humanoid ground projection and filtered according to navmesh connectivity, navmesh snap error, height consistency, and unobstructed line of sight. An independent half-cell-offset probe lattice is used to verify viewpoint coverage. Specifically, every retained close-stop probe that is both visible and navigable must lie within a geodesic distance of 0.15 m from at least one evaluation viewpoint, providing a 0.05 m margin below the 0.20 m success threshold.

Runtime integration. At runtime, the humanoid asset is instantiated at the episode goal pose before the first RGB-D observation is processed. Neither the goal pose nor the evaluation viewpoints are exposed to the navigation policy. Person detections are converted into RGB-D point clouds and projected into the same semantic goal map used for scene-native ObjectNav categories. The same planning procedure and STOP logic are applied to both scene-native and externally instantiated targets, without introducing a person-specific navigation branch.

E Physical Grounding and Scope

Scope of physics grounding. FireWorld is designed as a physics-grounded infrastructure for embodied navigation rather than an engineering-grade fire simulator. Here, *physics-grounded* refers to three aspects of the simulation: fire scenarios are parameterized by physically meaningful scene, material, and source attributes; the latent hazard state evolves through explicit physics-inspired mechanisms; and the resulting state is coupled to sensor observations using modality behaviors motivated by prior empirical studies. Table 2 summarizes these sources of physical and empirical grounding. The objective is to preserve the decision-relevant spatial and temporal structure of evolving fire hazards rather than to achieve CFD-level predictive accuracy.

Intended scope. This design places FireWorld between scripted or purely visual hazard augmentation and engineering-grade fire simulation. It provides an explicit, geometry-aware, and temporally evolving hazard state that can be consistently coupled to perception and planning while remaining computationally tractable for large-scale embodied evaluation. Accordingly, FireWorld is intended as a decision-oriented benchmark environment rather than a substitute for high-fidelity fire prediction.

F Baseline Details

F.1 Global Planning Baselines

Random Sampling (Random). Random Sampling selects long-term goals stochastically from navigable and reachable locations in the current map. This strategy does not explicitly consider exploration utility, navigation cost, hazard information, or inter-robot coordination.

Cooperative Greedy (Nearest / Fill) [29]. We consider two greedy frontier-selection strategies. The *Nearest-Frontier Greedy* (Nearest) assigns each robot to the reachable frontier with the minimum navigation cost, while the *Largest-Frontier Greedy* (Fill) selects the reachable frontier with the largest frontier region.

Cost–Utility (Co-UT) [7]. Cost–Utility balances frontier exploration utility against navigation cost. Each candidate frontier $f \in \mathcal{F}_t$ is scored as

$$S_{\mathrm{CU}}(f) = U(f) - \lambda_{\mathrm{CU}} C(f), \tag{15}$$

where $U(f)$ denotes the utility determined by frontier size, $C(f)$ represents the distance from the robot to the frontier, and λ_{CU} controls the trade-off between exploration utility and navigation cost. Each robot independently selects the frontier with the highest $S_{\mathrm{CU}}(f)$.

VLM-Based Planner. Beyond conventional frontier assignment strategies, we additionally consider recent representative navigation approaches [25, 39] as strong VLM-based baselines. These methods leverage vision-language models for high-level spatial and semantic reasoning and long-term navigation decisions. At each global planning step, the VLM receives the updated multi-modal prompt, which includes both semantic cues and hazard information, and assigns a long-term frontier goal to each robot. The assignment jointly considers two objectives: (i) maximizing exploration efficiency by reducing inter-robot redundancy and leveraging the potential semantic relevance between candidate frontiers and the target, and (ii) mitigating fire exposure by favoring lower-risk frontiers with higher perception confidence. The detailed multi-modal prompt design is provided in Appendix G.

Importantly, for all global planners, hazard awareness is incorporated as a soft preference rather than a hard exclusion criterion, since the target may lie within or beyond a hazardous region. Higher-risk frontiers therefore remain selectable when they exhibit strong semantic relevance or are necessary for continued exploration. If no preferable frontier is available, fallback goals are selected from explored regions with relatively low estimated risk.

F.2 Local Planning Baselines

Fast Marching Method (FMM) [24]. FMM computes a path to the assigned goal by propagating an arrival-time field over the navigable space. To incorporate fire risk, we modulate the propagation speed using the hazard map:

$$F(x) = \frac{1}{1 + \alpha H(x)}, \tag{16}$$

where $H(x)$ denotes the aggregated hazard level and α controls the trade-off between safety and efficiency. Higher-risk regions therefore induce slower propagation and larger traversal costs, biasing the resulting path toward safer areas. The local trajectory is obtained by following the negative gradient of the arrival-time field.

A* [5]. A* performs deterministic grid-based graph search from the robot's current position to the assigned goal using the same occupancy representation and traversability constraints. In the hazard-aware setting, its edge or cell costs are augmented using the same hazard representation employed by risk-aware FMM.

PointNav DD-PPO (PointNav) [33]. PointNav DD-PPO uses a frozen pretrained PointNav policy for local navigation. The assigned global goal is converted into the point-goal representation required by the policy, which then predicts navigation actions from its sensory observations. Because the pretrained policy does not receive hazard information, it is treated as a risk-blind learned planner.

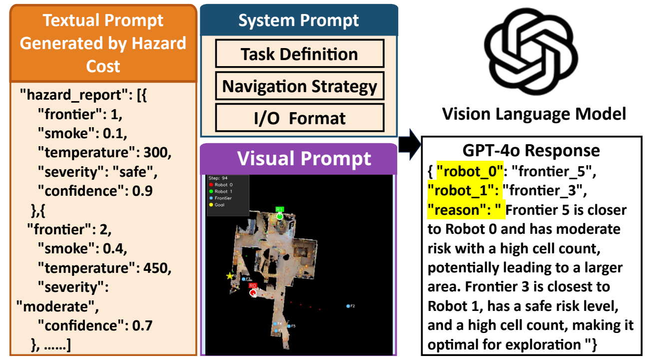

Figure 16: Multi-modal prompt design for VLM-based global planner.

G VLM-Based Global Planner

The VLM receives a compact yet expressive prompt consisting of a visual input and a textual input. An example of the multi-modal prompt is presented in Fig. 16.

Visual prompt. The visual input is a top-down global map rendered from the merged hazard-aware 3D point cloud and aligned with the 2D obstacle–hazard exploration map. Robot positions and identifiers are explicitly marked, while all

Figure 17: Unitree Go2 quadruped robots with an Intel RealSense D455 RGB-D camera and a Livox Mid 360 LiDAR.

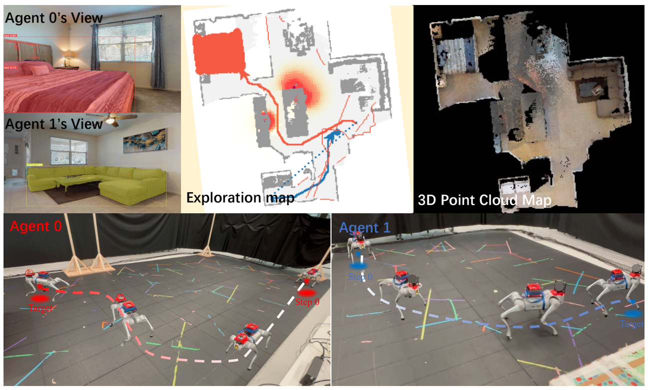

Figure 18: Real-world multi-robot navigation to find a bed. It shows the first-person RGB images, the exploration map, and the 3D point cloud map.

candidate frontiers are highlighted and labeled. This representation integrates geometric structure, semantic context, and multi-robot configuration for global spatial reasoning. Compared to prior VLM-based frameworks such as CoNavGPT [39] and MCoCoNav [25], which evaluate frontier clusters and assign a single waypoint per robot using separate visual inputs, FireNav uses a single aggregated visual input and performs one global reasoning step to generate frontier decisions for all robots, enabling coordinated long-horizon planning without sequential frontier inference.

Textual prompt. The textual input consists of a system prompt defining the fire-response navigation task and a frontier-level hazard report in JSON format. The system prompt specifies the mission objective, multi-robot setting, map semantics, and output constraints, while the hazard report summarizes estimated smoke density, temperature, hazard severity, and confidence for each frontier.

H Detailed Experimental Results

This section provides detailed experimental results supporting the evaluation presented in the paper. Table 4 compares different global–local planner combinations under hazard-free and dynamic-fire conditions for both ObjectNav and PersonNav. Table 5 reports the overall performance of FireNav under dynamic fire, covering task success, navigation efficiency, and hazard exposure. Table 6 examines the effect of different perception configurations on navigation performance and safety. Table 7 isolates the contribution of risk-aware planning by comparing risk-blind and risk-aware variants. Finally, Table 8 evaluates how navigation performance changes with the number of cooperating agents.

I Physical Experiments

This section provides additional details of the physical experiments used to validate the real-world feasibility of FireNav. We deploy the framework on two Unitree Go2 quadruped robots equipped with RGB-D and LiDAR sensors for cooperative target search in an indoor environment. Figs. 17 and 18 present the hardware platform and a representative navigation episode, illustrating the robots' first-person observations, shared exploration map, and reconstructed 3D point-cloud map. Table 3 further reports the runtime breakdown of the online navigation pipeline, including perception, mapping, planning, communication, and action execution.

Table 2: Sources of physical and empirical grounding in FireWorld. The benchmark targets decision-relevant physical structure rather than CFD-level predictive accuracy.

Stage	Grounding	FireWorld realization
Scene and material inputs	Physical attributes	Object geometry, structural boundaries, flammability, and smoke yield.
Fire initialization	Physical source conditions	Ignition location and time, source temperature, fuel mass, and intensity.
Hazard evolution	Physics-inspired mechanisms	Heat diffusion, buoyancy, ignition, fuel consumption, flame and smoke generation, dissipation, and cooling.
RGB-D and thermal sensing	Hazard-coupled observations	Smoke-dependent RGB-D degradation and temperature-derived thermal response.
Radar sensing	Empirically motivated behavior	Sparse geometric observations approximating the smoke-resilient characteristics of mmWave sensing.
Intended fidelity	Decision-oriented physical structure	Reproducible, geometry-aware, and temporally evolving hazards for embodied perception and planning.

Table 3: Runtime breakdown of the online navigation pipeline. Values are reported as the mean and standard deviation over 100 runs. VLM planning is invoked only during global replanning.

Statistic	Runtime (ms)						
	Preprocessing	Mapping	Exploration Planning	VLM Planning	Visualization	Communication	Action Completion
Mean	57.0	89.63	100.1	2155.4	33.7	4.31	378.4
Std.	32.6	34.16	15.9	578.6	12.1	13.83	112.43

Table 4: Performance comparison under hazard-free and dynamic-fire conditions. Higher SR and SPL are better, while lower DTG, NS, AHE, and SHER are better. AHE and SHER are reported only under dynamic fire. Bold and underlined values denote the best and second-best results, respectively, within each task and condition.

		Hazard-Free				Dynamic Fire				Safety under Fire	
Global	Local	SR ↑	SPL ↑	DTG (m) ↓	NS ↓	SR ↑	SPL ↑	DTG (m) ↓	NS ↓	AHE (%) ↓	SHER (/10k) ↓
ObjectNav											
Nearest	FMM	<u>58.3%</u>	.276	1.830	214.9	16.50%	.091	4.095	320.5	10.63	<u>.47</u>
Nearest	A*	37.0%	.220	3.063	295.7	21.00%	.121	3.701	309.9	11.60	1.37
Nearest	PointNav	49.3%	.281	2.069	264.8	36.50%	.231	2.515	239.9	9.55	2.92
Co-UT	FMM	56.2%	.259	1.982	<u>214.6</u>	17.00%	.087	4.332	333.5	9.37	.75
Co-UT	A*	35.7%	.188	2.922	298.5	20.50%	.107	4.296	320.0	10.66	.78
Co-UT	PointNav	53.0%	<u>.286</u>	1.945	246.4	41.00%	.238	2.538	226.2	8.95	3.76
Fill	FMM	**59.7%**	.285	<u>1.623</u>	**197.1**	20.50%	.114	3.920	311.7	9.61	.56
Fill	A*	35.7%	.194	3.169	293.9	20.50%	.111	3.591	296.2	8.98	1.27
Fill	PointNav	50.8%	.284	1.917	244.9	<u>43.00%</u>	<u>.257</u>	<u>1.928</u>	**219.3**	8.33	3.31
Random	FMM	51.0%	.228	2.409	249.7	16.50%	.076	3.852	314.0	10.84	.56
Random	A*	35.7%	.159	2.996	314.0	23.00%	.097	3.602	315.0	9.39	1.11
Random	PointNav	36.2%	.215	2.698	298.2	35.50%	.199	2.431	287.5	<u>7.40</u>	.61
GPT-4o†	FMM	58.0%	.264	2.044	222.6	26.80%	.164	2.964	291.89	12.6983	**.3532**
GPT-4o†	A*	36.5%	.200	3.497	299.7	30.77%	.160	2.838	276.82	**6.5875**	1.1578
GPT-4o†	PointNav	58.0%	**.314**	**1.575**	240.5	**49.00%**	**.294**	**1.859**	<u>221.95</u>	8.1944	1.6896
PersonNav											
Nearest	FMM	46.5%	.279	6.658	294.1	11.50%	.082	8.473	335.1	15.42	1.04
Nearest	A*†	25.0%	.154	8.446	366.9	17.69%	.124	8.094	338.4	15.81	1.31
Nearest	PointNav	37.8%	.252	6.878	332.8	46.00%	.330	6.500	266.7	12.85	2.44
Co-UT	FMM	48.5%	.269	6.195	287.8	13.00%	.094	8.453	348.1	15.89	<u>.65</u>
Co-UT	A*	25.5%	.162	8.172	364.9	16.50%	.118	8.765	342.8	16.34	1.53
Co-UT	PointNav	42.0%	.260	6.340	302.2	<u>50.50%</u>	**.353**	<u>6.221</u>	<u>238.3</u>	<u>11.64</u>	2.83
Fill	FMM	52.5%	.287	**5.234**	276.8	13.00%	.095	8.482	334.5	14.68	.82
Fill	A*†	28.2%	.175	7.765	360.6	8.84%	.063	8.536	376.0	16.86	1.00
Fill	PointNav	51.2%	.305	<u>5.349</u>	<u>272.4</u>	49.50%	.340	6.241	253.3	11.98	2.37
Random	FMM	47.8%	.266	5.930	285.8	17.00%	.109	8.367	338.7	13.39	.89
Random	A*	30.5%	.168	7.162	356.4	16.00%	.106	8.113	330.2	14.75	2.50
Random	PointNav	40.0%	.250	6.413	315.6	43.00%	.312	7.341	287.4	12.21	1.20
GPT-4o†	FMM	<u>57.5%</u>	<u>.359</u>	6.205	274.3	19.75%	.126	10.557	351.15	15.5696	**.1758**
GPT-4o†	A*	22.5%	.154	8.696	382.5	13.21%	.126	12.025	304.53	**11.1868**	1.8587
GPT-4o†	PointNav	**59.0%**	**.380**	5.688	**264.2**	**54.00%**	<u>.346</u>	**6.187**	**227.64**	14.7305	3.1849

Table 5: Overall performance of FireNav under dynamic fire conditions. Results for ObjectNav (Obj.) and PersonNav (Per.) are shown as paired subcolumns under each metric. Higher SR and SPL are better, while lower DTG, NS, Average Hazard Exposure (AHE), and Severe Hazard Exposure Rate (SHER) are better. Bold and underlined values denote the best and second-best results for each task, respectively.

Global Planner	Local Planner	SR ↑		SPL ↑		DTG (m) ↓		NS ↓		AHE (%) ↓		SHER (/10k) ↓	
		Obj.	*Per.*	*Obj.*	*Per.*	*Obj.*	*Per.*	*Obj.*	*Per.*	*Obj.*	*Per.*	*Obj.*	*Per.*
Nearest	FMM	.420	.325	.178	.187	2.672	7.698	272.8	303.8	9.0355	13.1070	<u>.3327</u>	.8384
Nearest	A*	.447	.3871	.224	.240	2.486	7.421	258.0	273.8	9.8600	13.4385	1.0938	1.0688
Nearest	PointNav	.480	.385	.266	.255	2.317	7.066	258.9	299.3	8.1175	10.9225	2.3140	1.9885
Co-UT	FMM	.395	.370	.164	.224	2.958	7.597	282.4	307.1	7.9645	13.5065	.5519	<u>.5105</u>
Co-UT	A*	.429	.4219	.219	.266	2.574	7.227	<u>235.6</u>	275.0	9.0610	13.8890	.6652	1.2090
Co-UT	PointNav	.515	.405	<u>.277</u>	.247	2.273	6.986	247.4	289.3	7.6075	<u>9.8940</u>	2.9841	2.2709
Fill	FMM	.330	.320	.159	.175	2.773	7.843	260.7	313.5	8.1685	12.4780	.4795	.6193
Fill	A*	.448	.3802	.214	.243	2.343	7.639	250.8	292.0	7.6330	14.3310	1.0200	.8284
Fill	PointNav	.475	.410	.258	.252	<u>2.048</u>	<u>6.585</u>	242.5	281.1	7.0805	10.1830	2.6021	1.8895
Random	FMM	.350	.360	.153	.197	3.109	7.606	272.7	292.1	9.2140	11.3815	.4880	.6871
Random	A*	.397	.3593	.189	.209	3.028	7.304	257.5	<u>273.4</u>	7.9815	12.5375	.8755	1.9943
Random	PointNav	.385	.345	.224	.233	2.767	7.492	282.1	300.5	<u>6.2900</u>	10.3785	.4932	.9319
GPT-4o	FMM	<u>.550</u>	<u>.535</u>	.254	<u>.336</u>	2.136	6.640	**229.53**	281.99	10.7936	13.2342	**.3268**	**.0887**
GPT-4o	A*	.360	.215	.196	.151	3.567	9.029	305.69	390.15	**5.5994**	**9.5088**	.8996	1.4738
GPT-4o	PointNav	**.570**	**.585**	**.312**	**.377**	**1.603**	**5.738**	245.31	**269.49**	6.9652	12.5209	1.3249	2.5048

Table 6: Ablation of multi-modal perception and fusion in fire-scene ObjectNav. FMM is fixed as the local planner. Results are averaged over the matched risk-blind and oracle-aware settings. Higher SR and SPL are better, while lower DTG, NS, AHE, and SHER are better.

		Navigation Performance				Safety	
Global	Local	SR ↑	SPL ↑	DTG (m) ↓	NS ↓	AHE (%) ↓	SHER ($\times 10^{-3}$) ↓
RGB-D Only							
Nearest	FMM	.190	.103	3.552	288.35	8.275	.0478
Co-UT	FMM	.190	.101	3.793	307.25	8.820	.0613
Fill	FMM	.215	.122	3.647	300.85	7.885	.0373
Random	FMM	.183	.084	3.587	318.40	7.890	.0510
GPT-4o	FMM	.200	.114	4.129	319.45	9.475	.0390
Multi-modal Perception and Fusion							
Nearest	FMM	.375	.176	2.781	260.65	8.830	.2060
Co-UT	FMM	.385	.181	2.929	248.75	7.940	.2383
Fill	FMM	.408	.195	2.557	243.65	8.165	.2669
Random	FMM	.318	.145	3.231	266.65	8.655	.3140
GPT-4o	FMM	.353	.173	3.193	280.05	8.840	.1739

Table 7: Ablation of risk awareness in fire-scene ObjectNav. Higher SR and SPL are better, while lower DTG, NS, Average Hazard Exposure (AHE), and Severe Hazard Exposure Rate (SHER) are better. AHE is reported as a percentage, and SHER is reported in units of 10^{-3}.

		Navigation Performance				Safety	
Global	Local	SR ↑	SPL ↑	DTG (m) ↓	NS ↓	AHE (%) ↓	SHER ($\times 10^{-3}$) ↓
Risk-Blind							
Nearest	FMM	.375	.232	7.617	310.105	13.22	.306
Fill	FMM	.365	.228	7.524	304.095	12.52	.345
Co-UT	FMM	.360	.227	7.643	293.705	12.61	.460
Random	FMM	.350	.179	7.893	285.905	12.42	.752
GPT-4o	FMM	.485	.234	2.482	239.3	11.0258	.2716
Risk-Aware							
Nearest	FMM	.420	.178	2.672	272.78	10.8201	.09165
Fill	FMM	.330	.159	2.773	260.72	9.6151	.22055
Co-UT	FMM	.395	.164	2.958	282.40	**8.8417**	.13279
Random	FMM	.350	.153	3.109	272.66	9.5813	.29341
GPT-4o	FMM	**.550**	**.254**	**2.136**	**229.53**	10.7936	**.03268**

Table 8: Navigation performance with different numbers of agents under dynamic fire conditions. FMM is used as the local planner for all configurations.

# Agents	Global Planner	SR ↑	DTG (m) ↓	SPL ↑	NS ↓	AHE (%) ↓	SHER (/10k) ↓
1	Nearest	35.0%	3.200	0.145	292.4	9.71	0.34
	Co-UT	34.0%	3.180	0.142	287.6	9.37	0.92
	Fill	36.5%	2.940	0.155	278.2	10.83	1.52
	Random	26.5%	3.800	0.126	329.0	10.20	2.30
	Average	33.00%	3.280	0.142	296.8	10.03	1.27
2	Nearest	42.0%	2.672	0.178	272.8	10.82	0.92
	Co-UT	39.5%	2.958	0.164	282.4	8.84	1.33
	Fill	33.0%	2.773	0.159	260.7	9.62	2.21
	Random	35.0%	3.109	0.153	272.7	9.58	2.93
	Average	37.38%	2.878	0.164	272.1	**9.71**	1.84
3	Nearest	41.5%	2.449	0.215	241.4	10.32	1.31
	Co-UT	46.5%	2.096	0.225	221.2	10.11	1.28
	Fill	47.5%	2.006	0.233	227.7	9.54	1.61
	Random	34.5%	2.802	0.155	259.0	9.77	1.80
	Average	**42.50%**	**2.338**	**0.207**	**237.3**	9.93	1.51
4	Nearest	40.5%	2.647	0.206	236.5	8.72	1.37
	Co-UT	46.0%	2.263	0.247	218.3	10.38	1.15
	Fill	43.5%	2.204	0.221	237.9	10.15	1.05
	Random	31.0%	2.860	0.151	257.5	10.35	1.80
	Average	40.25%	2.493	0.206	237.5	9.90	**1.36**